

Sai Jagadeesh Muralikrishnan

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EDUCATION

University of Maryland

Master of Engineering in Robotics — CGPA: 3.8/4.0

- Relevant Coursework: Control Systems, Machine Learning, Computer Vision, Perception in Robotics

Rajalakshmi Engineering College

Bachelor of Engineering in Mechatronics — CGPA: 8.7/10.0

- Relevant Coursework: Embedded systems, Controls, Power electronics

College Park, MD

Aug. 2023 – May 2025

Chennai, India

Aug. 2018 – Jul. 2022

EXPERIENCE

Robotics Software Engineer

Sep. 2025 – Present

Miso Robotics

Pasadena, CA

- Developed robotic arm behaviors for Flippy product lines using ROS 2 and python for defining cooking tasks.
- Working on motion planning algorithms for constrained environments, achieving 100% success in state transitions.
- Designed HIL test suites for serial 6-DOF arm to validate calibration behaviors and feature integrity.
- Refined site-specific configuration files and simulations to eliminate planning failures and reduce system downtime.

Robotics Engineering Intern

May 2024 – Aug. 2024

Kick Robotics

Bethesda, MD

- Reduced QA cycle time by 20% by automating CI/CD pipelines (ROS 2, Docker, PyTest, GoogleTest)
- Decreased system latency by 15% by refactoring ROS 2 (C++) nodes and optimizing system integration on Jetson
- Increased depth-map precision by 12% via field calibration of Basler ToF and Intel RealSense D435 cameras for outdoor detection applications.

Embedded Systems Engineer

Oct. 2022 – Jul. 2023

TuTr Hyperloop

Chennai, India

- Reduced processing latency by 25% by building sensor fusion pipelines in C++/Python, integrated via MATLAB
- Achieved 99.9% system uptime by designing Vehicle Control Unit (VCU) signaling over CAN protocols
- Increased test coverage by 40% through automated Git-based pipelines for unit testing and HIL validation

PROJECTS

CareBotix - AI Patient Monitoring | YOLOv8, OpenCV, PyQt6, Flask, MongoDB

Apr. 2025 – Apr. 2025

- Designed AI-based patient monitoring with YOLOv8-pose detection, won Best Health Track Project Award
- Built full-stack architecture using PyQt6 GUI and Flask backend with MongoDB and REST APIs

Text-to-Command Robot Navigation | ROS 2, Transformers, LoRA, PyTorch, NLP

Nov. 2024 – Dec. 2024

- Fine-tuned T5-Small with LoRA adapters on 24.5K instructions, achieving 98.5% sequence accuracy for language-based mobile robot navigation
- Integrated trained model with ROS 2 and Gazebo; optimized inference performance by evaluating latency and model size tradeoffs

Multi-Agent Robotic Exploration | ROS 2, C++, Gazebo, OpenCV, Open3D, PyQt5

Nov. 2023 – Dec. 2023

- Designed and validated Monte Carlo Tree Search (MCTS) path planning through 200+ simulations.
- Built visualization system using OpenCV/Open3D with PyQt5 dashboard and managed with repository

TECHNICAL SKILLS

Languages: Python, C++, Bash, MATLAB, Shell Scripting, Linux

Frameworks & Platforms: ROS 2, PyTorch, TensorFlow, Flask, Gazebo, Isaac Sim, CUDA

Tools & DevOps: Git, GitHub, GitLab, Docker, AWS, GCP, Jira, CI/CD, Unit Testing, Debugging, REST APIs

Libraries: OpenCV, YOLOv8, PCL, Open3D, NumPy, Transformers, Nav2, MoveIt, CAD/CAE, Statistics

Other: Data Structures, SolidWorks, Cross-functional Collaboration, Technical Presentations

NOTABLE ACHIEVEMENTS

Hackathons: 3rd Place Winner and Best Health Track Award for CareBotix at Morgan Hacks 2025

Scholarships: Pathways to PhD Scholarship, Pathways to Profession Scholarship from Maryland Robotics Center (2024)

Leadership: President of COSMO (Mechatronics Department Club), REC (Apr 2021 – Apr 2022)

Awards: Best Paper Award (3rd Place) – International Conference on Data Science, ML & Applications 2021

Publications: "Wireless Animatronic Hand Using Infrared Sensor" – ICDSMLA 2021, Springer Nature Singapore

Competition: Best Product Analyst Award from Designer's Consortium, REC (2020)